

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, ML & Distributed Systems Aug 2025 to May 2026

- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Built and exercised **14 interoperability scenarios** covering networking, dependency, and **failure recovery** behavior in a distributed real-time system.
- Boosted malware classification to **80.2%** accuracy across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.

Turito | Software Engineering Intern, Real-Time Infrastructure Jun 2022 to Oct 2022

- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Prototyped a **Rust** + **QUIC** media path that supported about **8,000** lightweight subscribers with about **24%** lower startup time.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Surfaced backend fixes that raised equivalent-load signaling throughput by about **31%**.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, TypeScript, SQL

AI Infrastructure: LLM serving gateways, semantic caching, HNSW, vector search, PyTorch, Hugging Face, sentence-transformers, OpenAI and Anthropic APIs

Systems & Performance: AVX2/SIMD, int8 quantization, profiling, concurrency, Tokio, asyncio, Linux, CMake, pybind11

Backend & Infra: FastAPI, gRPC, Kafka, Temporal, Redis, PostgreSQL, SQLite, Docker, AWS, SSE, QUIC

PROJECTS

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.
- Served **78%** of requests from cache and cut modeled provider cost by **76%**, with live p50/p95, token, and cost metrics.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnswlib and **2.5x** faster than FAISS on 100K vectors.
- Cut traversal memory by **75%** (51 MB to 12.8 MB) with **int8** traversal on AVX2 kernels plus exact float re-ranking, and cache-aware node reordering.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, GPU Programming, Machine Learning

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)